

Towards a Theory of Simplicity in Agile Software Development: A Qualitative Study

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Abstract—Context: Several research works emphasise that the concept of simplicity is, in itself, by far not a simple concept, mainly because there are many perspectives on the perception of simplicity. **Purpose:** To understand how project managers and software engineers interpret their experiences in agile software projects considering the simplicity issues in agile software development. **Method:** Semi-structured interviews were carried out with practitioners within a software development company. The data was analysed using grounded theory techniques. **Results:** A set of categories (lightweight process, knowledge acquisition, personal communication, time-consuming, and product with value) that affect the simplicity in agile software development were extracted. Finally, the categories and propositions with higher explanatory power were used to create a provisional theory about simplicity in agile software projects. **Conclusion:** The results show that a better understanding of the implications of simplicity on agile software development may contribute to the projects' successes.

Keywords – Simplicity, Agile Software Development, Qualitative Study.

I. INTRODUCTION

Considering the emerging challenges and opportunities in software development, organisations have been changing from a hierarchical approach of project management to being more collaborative and flexible. The need for distribution of responsibility and initiative to support adaptation to change is a familiar territory of agile approaches [1].

In line with the agile manifesto, Agile Software Development (ASD) has proven to be an important set of methods in promoting simplicity issues. Although there is a variety of methodologies and frameworks of ASD, few academic studies directly address simplicity [2][3][4]. This study aims to provide the preliminary results of an on-going research focused on understanding how project managers and software engineers interpret their experiences about simplicity in agile software development in the workplace.

To find answers to the research question, a qualitative case study in a medium software company in England was conducted, in which practitioners were interviewed to collect data about their perception in practice regarding simplicity phenomena.

II. BACKGROUND

The study of simplicity is an interdisciplinary endeavour with many dimensions, concepts and attributes [3], as described in the following subsections.

A. Simplicity in Information and Communications Technology

The study of Simplicity in IT development dates back to Margaria and Steffen [5]. Based on a systematic literature review and direct interaction with IT experts, Margaria and Steffen, with Bosselmann and Floyd, [2][3] compiled a set of dimensions of simplicity for possible lines of action: the art of knowing, structure, orthogonality, size, transparency, communication, automation, context and subjectivity/felt complexity. Their reasoning and examples show that such dimensions rarely occur or are acted upon in isolation, so that in reality several dimensions play a role as initiator, mediator, or target of a simplicity-driven change. This understanding seems to suggest that agile software development should be an ideal turf for embracing simplicity.

B. Simplicity in Agile Software Development

In order to satisfy the agile values, some principles [6] should be respected, including the principle which states that “Simplicity, the art of maximising the amount of work not done – is essential”. In essence, agile methods emphasise simplicity. The goal is to get user feedback quickly by delivering software at short increments, even if it covers only a subset of the expected functionality [4].

Meyer [4] affirms that who has ever obtained an initial solution to a problem of any kind, found it complex and tried to simplify it. Achieving simplicity often means adding work, sometimes lots of it. As observed by Margaria, Steffen, Bosselmann and Floyd [2][3], the community of researchers and practitioners believes that the philosophy of simplicity is strategically important, yet still insufficiently understood.

This research work provides empirical evidence about simplicity at work and leads towards proposing a first model of simplicity in agile software development that can help researchers and practitioners to better understand the benefits of simplicity in this context.

III. METHOD

This research adopts the roadmap for building theories from case study proposed by Eisenhardt [7]. According to Eisenhardt, without a research focus, it is easy to become overwhelmed by the volume of data. In this sense, the research focus is defined by the following initial research question: *How is simplicity understood by the agile team members?*

Semi-structured interviews were performed with experienced practitioners. The interview script was composed of open-end questions (Fig. 1), structured as (i) respondent demographic profile, (ii) simplicity in agile software development, and (iii) key dimensions of simplicity. Initially (see Q5 and Q6), the interview guide encompasses quick questions, aimed at exploring the experience and the background of the participants. The next phases were presented in a funnel model, beginning with general questions focused on understanding the broad aspects of simplicity in agile software development (see Q12), which were then refined toward more specific questions.

...
Q5. How many years have you participated in or were involved in initiatives using agile methods?
Q6. Please indicate the agile methodologies in which you work (or worked).
...
Q12. What do you understand by simplicity in agile projects?
...
Q21. How would you describe the importance of "knowledge" as dimension that lead to simplicity in the context of agile software development? Please indicate some examples.
Q22. How would you describe the importance of "communication" as dimension that lead to simplicity in the context of agile software development? Please indicate some examples.
Q23. How would you describe the importance of "time" as dimension that lead to simplicity in the context of agile software development? Please indicate some examples.
Q24. Which other Dimension do you believe that lead to simplicity in the context of agile software development?

Fig. 1. Interview guide extract

The general questions encouraged relevant and unbiased reflections bringing more details when answering the specific questions. Some questions (see Q21, Q22 and Q23) were based on the dimensions that lead to simplicity identified by Margaria, Steffen, Bosselmann and Floyd [2][3]. The last question of our instrument (see Q24) explores additional possibilities of dimensions that lead to simplicity in agile software development.

The participants were interviewed at their workplace. The data was collected over seven weeks in November and December of 2015 and January of 2016. We used ATLAS.ti¹ tool to analyse and synthesise the data, which were labelled thorough qualitative codes to distill, identify similarities and sorts them to describe the phenomenon of simplicity in agile software development.

In sequence, the constant comparative method of qualitative analysis [8] was adopted to compare each code from the same interview and those from other interviews. As we continuously compared the codes, many fresh concepts emerged. As

the process of data analysis progressed, relationships among categories and memos were written to keep us involved with the analysis, thus helping to increase the level of abstraction of our ideas about the codes, concepts, categories and possibly even relationships. These memos forced us to look beyond impressions and see evidence through multiples lenses.

IV. RESULTS

This section provides the contextual information regarding the selected software company and the participants involved in the study. In the sequel, the results relating the characterisation of simplicity, as well the central story that explains simplicity in agile software development are described.

1) *The Software Company*: It is a UK-based software development company, founded in 2000, with offices in London and Bristol. By the time of this study, the company had about 110 employers. The company is classified as a for-profit small organisation².

2) *The Participants*: Six skilled practitioners with different roles were purposely sampled, to achieve maximum variation in data collection. Table I compiles the participants' demographic profile. Due to anonymity and ethical issues, the participants are labelled by *P1* to *P6* codes.

TABLE I
PROFILE OF PARTICIPANTS

	Role	Education	Experience (years)	Agile
P1	Software Engineer Scrum Master	BSc	6 to 10	Scrum; Kanban
P2	Project Manager Scrum Master	MSc MBA	6 to 10	Scrum; Kanban XP; LSD; FDD
P3	Project Manager Scrum Master; Developer	BSc	6 to 10	Scrum; Kanban Lean
P4	Software Engineer Scrum Master; Test Lead	MSc	6 to 10	Scrum; Kanban Lean
P5	Software Engineer Designer	BSc	6 to 10	Scrum; Kanban LSD
P6	Software Engineer Scrum Master	BSc	6 to 10	Scrum; Kanban

A. Characterising Simplicity in ASD

Participants were asked about their understanding of simplicity in ASD. The findings of this step are presented here. For each quote, the following format was adopted: [*P participant number*]. From the agile team's perspective, simplicity in ASD is broadly related to different kind of categories, such as (i) **lightweight process**, (ii) **knowledge acquisition**, (iii) **time-consuming**, (iv) **product with value**, and (v) **personal communication**. These categories are better discussed in the following excerpts.

According to participants, simplicity is broadly related to **lightweight process**. It means that the overhead of the process is kept as small as possible, to maximise the amount of productive time available for getting useful work done.

"I would say that simplicity in agile is about not having too much complicated process." [P4]

¹ATLAS.ti website: <http://atlasti.com>

²European Union Commission Recommendation: <https://goo.gl/u0IPLr>

"It has to be straightfull to do. If it's a process, it hasn't to take a lot of different steps." [P5]

"(...) suppose like if an agile method is simple, there won't be too much processes involved." [P3]

When the team prioritise things that add knowledge at the beginning of the project, they could speed up the pace of understanding the requirements and therefore lower the risks of misunderstanding. Participants emphasised that **knowledge acquisition** is widely related to simplicity.

"Oh, I think there's probably two aspects to that (simplicity in ASD). One is that something has to be easy to understand (...)" [P5]

Essentially, the agile team needs to start by understanding their readiness for project's context and focusing resources most effectively to minimise **time-consuming** (time spent on process overhead). According to participants, time management also influences simplicity to avoid unnecessary time-consuming and downtime, as seen in the following quotes:

"it has to be quick to do." [P5]

"(...) suppose if an agile method is simple and it won't take me long. So it will be quick for me to use and either I won't do the wrong thing." [P3]

Fundamentally, agile teams must ensure consistent delivery of a product with value to customers. It makes sure that a product or service always is of the highest quality possible [9]. According to participants, focusing on **product with value** leads to simplicity, as seen in the following quote:

"That's (simplicity in ASD) about focusing on the few things that have the most impact. Kind of focusing on what the project is meant to achieve." [P4]

"Yeah, it is getting down to what is the client optimal tools and follow one of those, and one of the simplest steps we can imagine with the least amount of work we can do to get that attempt done." [P6]

In particular, the agile methodologies focus more on effective communication, collaboration, and coordination within a dynamic team environment than up-front planning and documentation [6]. According to evidence, **personal communication** with team and customer also promotes simplicity, as seen in the following quote:

"(simplicity in ASD) is also about focusing on things like communicating well with your team, with your customer." [P4]

On the other hand, participants also express their concern about the statement of simplicity. P5 declares that the principle of simplicity is itself not simple - it is a convoluted definition.

"(...) it's kind of a convoluted way of putting it. It's not a simple way of defining simplicity. It's not a simple definition." [P5]

B. Relating Practices and Building Propositions

The relationships among categories that emerged from the data analysis were organised in propositions, each one describing a particular view of the phenomenon, resulting in four key relationships.

Proposition 1: The inclination towards **lightweight process** enhances **personal communication**, reduce **time-consuming** tasks, and leads to **simplicity in ASD**. In general, agile methods are very lightweight processes that embody less process, employ short iteration cycles; actively involve users to establish, prioritise, and verify requirements; and rely on tacit knowledge within a team as opposed to documentation [9]. Participants identified that enhancement of personal communication and reduction of time-consuming are caused by the promotion of lightweight process.

"(...) focusing on the few things that have the most impact. Things like communicating well with your team, with your customer." [P4]

Proposition 2: The encouragement to **knowledge acquisition** promotes **simplicity in ASD**. The agile development environment is considered as a platform for the extraction of knowledge without extra effort, overcoming cultural and psychological barriers [10]. Participants identified knowledge acquisition as an important tool that leads to simplicity in ASD and helps the longer term productivity and flexibility of the team, as seen in the following quote:

"(...) if your knowledge that comes out is really useful, that certainly makes things simpler." [P6]

Proposition 3: The adoption of a **personal communication** with customer and agile team positively impact the **product with value** and leads to **simplicity in ASD**. The personal communication among the team members and customers were highlighted as crucial to achieve simplicity in ASD and ensure effective feedback, as seen in the following quote:

"(...) communication is definitely on the top of the pyramid of customer involvement" [P2]

Proposition 4: The optimisation of **time-consuming** positively promotes on development of **product with value**, and leads to **simplicity in ASD**. According to participants, the high-quality of product (value to the customer) is a consequence of a focused time prioritisation towards a thoughtful reduction of unnecessary work done upfront.

"(when) simpler, then each story will take less time to develop. They'll take less time to test, they'll take less time to understand the requirements" [P2]

C. Building the Provisional Theory

The propositions presented in the previous section were combined to build the central story that explains a model of simplicity in agile software development. The relationships between the categories were analysed and focused using selective coding [11] to provide the overall theoretical picture. In this instance, the analysis showed that **lightweight process** emerged as the fulcrum category of the study. This central relation is presented and illustrated in the model in Figure 2.

Furthermore, analysing the interrelationship between these categories, we find that there are two clusters of goals, organised in externally-oriented simplicity goals and internally-oriented simplicity goals. The first cluster is composed

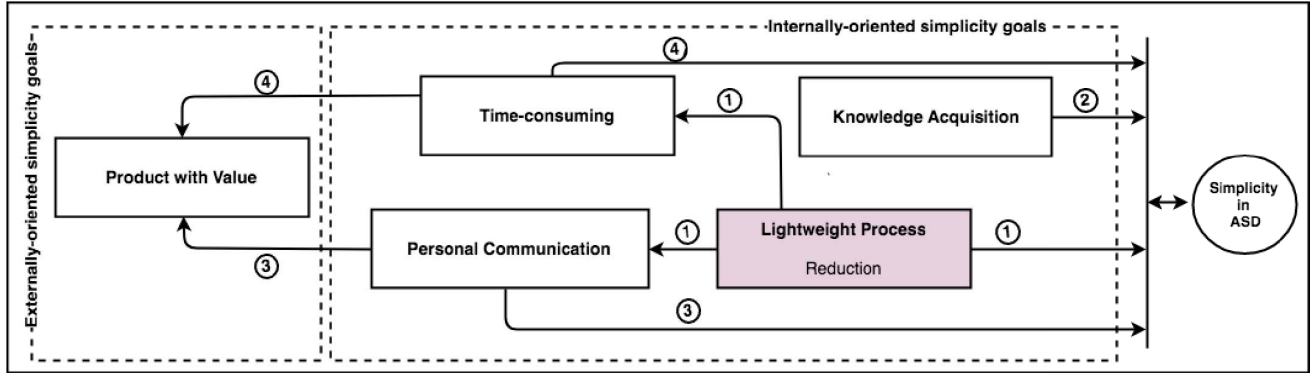


Fig. 2. Provisional Theory (Central Story of Simplicity in Agile Software Development)

by **lightweight process**, **knowledge acquisition**, **time-consuming**, and **personal communication**. The last two aim for **product with value** as externally-oriented simplicity goal. This structure also grouped internal and external goals of the agile manifesto.

V. DISCUSSION

In this section, we discuss the limitations and the validity of our results from the perspective proposed by Merriam [12].

Construct validity: Although we constantly compared and contrasted our construct definitions with the literature, a member checking can be conducted in the future to ensure if our interpretations were consistent with those of the participants.

Internal validity, or credibility: To increase credibility, we sampled participants with different roles in software projects, as described in Table I. The preliminary results were discussed between the authors to refine the categories and the propositions. A limitation is that we used only interviews as data collection method.

Finally, a common challenge in qualitative studies is to reach *theoretical saturation*. In this study, we interviewed six participants with different points of view and perceptions about the studied phenomenon. Although they contributed to a rich description of the phenomenon, we aim to replicate our protocol in other organisations.

VI. CONCLUSION AND FUTURE WORKS

In this article, we presented the main results of a qualitative study focused on understanding how practitioners interpret simplicity in agile software development projects and how these interpretations shape their work towards simplicity. This qualitative study was performed in a small software company in England that had just over 110 skilled practitioners.

Qualitative research techniques were applied to collect, analyse and synthesise data via interviews. Qualitative coding techniques were employed to identify the categories that lead to simplicity in agile software development. From the interpretation of data, we constructed propositions that describe the relationships, and also built a provisional theory that explains the phenomena.

In summary, the obtained results show that simplicity is aligned with the grounded categories of lightweight process, knowledge acquisition, time-consuming, personal communication, aiming for a product with value. The next phase of the investigation will apply the protocol in other organisations and use the member checking technique to check the consistency of our interpretations and to raise the theoretical level.

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REFERENCES

- [1] D. J. Fernandez and J. D. Fernandez, "Agile Project Management - Agilism Versus Traditional Approaches," *Journal of Computer Information Systems*, vol. 49, pp. 10–17, 2008.
- [2] B. D. Floyd and S. Bosselmann, "ITSy - Simplicity Research in Information and Communication Technology," *Computer*, vol. 46, no. 11, pp. 26–32, Nov. 2013.
- [3] T. Margaria, B. Steffen, and B. D. Floyd, "ITSy - Recommendation Document," University of Postdam, Postdam, Tech. Rep., 2011.
- [4] B. Meyer, *Agile! The Good, the Hype and the Ugly*. Zurich, Switzerland: Springer, 2014.
- [5] T. Margaria and B. Steffen, "Simplicity as a Driver for Agile Innovation," *Computer*, vol. 43, no. 6, pp. 90–92, Jun. 2010.
- [6] K. Beck and Beedle, "Manifesto for Agile Software Development," 2001, <http://agilemanifesto.org>. Accessed on January 27, 2017. [Online]. Available: <http://agilemanifesto.org>
- [7] K. M. Eisenhardt, "Building theories from case study research," *The Academy of Management Review*, vol. 14, no. 4, pp. 532–550, 1989.
- [8] B. Glaser, "The Constant Comparative Method of Qualitative Analysis," *Social Problems*, vol. 12, no. 4, pp. 436–445, 1965.
- [9] A. Cockburn, *Agile Software Development*. Addison-Wesley Professional, 2001.
- [10] M. Levy, D. Telekom, L. Ben, and T. Israel, "Knowledge Management in Practice : The Case of Agile Software Development Orit Hazzan Department of Education in Technology and Science," pp. 60–65, 2009.
- [11] A. L. Strauss, *Qualitative Analysis for Social Scientists*. San Francisco, CA, United states: Cambridge University Press, 1987.
- [12] S. B. Merriam, *Qualitative Research: A Guide to Design and Implementation*, 2nd ed., ser. Jossey-Bass higher and adult education series. United States of America: John Wiley & Sons, 2009.